Here's a **comprehensive list of AI libraries**, categorized by different AI domains. I've included popular and widely used libraries across **Machine Learning (ML), Deep Learning (DL), Natural Language Processing (NLP), Computer Vision (CV), Generative AI, Reinforcement Learning (RL), Multi-Modal AI, and Specialized AI**.

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1. Machine Learning (ML)

Used for: Regression, classification, clustering, dimensionality reduction, ensemble methods, etc.

- **Scikit-learn** (Python) – ML algorithms, preprocessing, metrics
- **XGBoost** (Python, C++) – Gradient boosting for structured data
- **LightGBM** (Python, C++) – Fast gradient boosting
- **CatBoost** (Python, C++) – Gradient boosting for categorical features
- **MLlib** (Apache Spark, Scala/Java/Python) – Big data ML
- **H2O.ai** (Java, Python, R) – Scalable ML
- **Orange3** (Python) – Visual ML workflow
- **Weka** (Java) – ML algorithms and visualization

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2. Deep Learning (DL)

Used for: Neural networks, CNNs, RNNs, LSTMs, Transformers

- **TensorFlow** (Python, C++) – DL and neural networks
- **Keras** (Python) – High-level API for TensorFlow
- **PyTorch** (Python, C++) – DL, research-friendly
- **MXNet** (Python, Scala, R, C++) – Scalable DL
- **Chainer** (Python) – Flexible neural networks
- **Theano** (Python, legacy) – Symbolic tensor computation
- **DL4J** (Java) – Deep learning in Java

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3. Natural Language Processing (NLP)

Used for: Text analysis, language understanding, chatbots, sentiment analysis

- **NLTK** (Python) – Text preprocessing, tokenization, corpora
- **spaCy** (Python) – Efficient NLP pipelines
- **Gensim** (Python) – Topic modeling, word embeddings
- **Transformers (HuggingFace)** (Python) – Pretrained BERT, GPT, T5
- **Stanford CoreNLP** (Java, Python wrapper) – POS tagging, parsing
- **OpenNLP** (Java) – NLP tasks
- **TextBlob** (Python) – Simple NLP tasks
- **AllenNLP** (Python) – Research in NLP

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4. Computer Vision (CV)

Used for: Image classification, object detection, segmentation

- **OpenCV** (C++, Python, Java) – Image/video processing
- **Pillow/PIL** (Python) – Image manipulation
- **scikit-image** (Python) – Image processing
- **Detectron2** (Python, PyTorch) – Object detection, segmentation
- **YOLO (Ultralytics)** (Python) – Real-time object detection
- **MediaPipe** (Python, C++) – Face, hand, pose detection
- **SimpleCV** (Python) – Computer vision framework

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5. Generative AI

Used for: Image generation, music, text, video, 3D models

- **GANs (TensorFlow/PyTorch)** – Generative Adversarial Networks
- **StyleGAN / StyleGAN2** – High-quality image generation
- **DALL·E / OpenAI API** – Text-to-image generation
- **Stable Diffusion** – Text-to-image generation
- **VQ-VAE / VQGAN** – Image synthesis
- **MusicVAE** (TensorFlow) – Music generation
- **DeepDream** (TensorFlow) – Visual pattern generation

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6. Reinforcement Learning (RL)

Used for: Training agents, robotics, games, control systems

- **OpenAI Gym** (Python) – RL environments
- **Stable Baselines3** (Python) – RL algorithms
- **RLlib (Ray)** (Python) – Scalable RL
- **TensorForce** (Python) – RL framework
- **Dopamine (Google)** – RL research framework
- **Keras-RL** (Python) – RL with Keras
- **Unity ML-Agents** (C#, Python) – Game-based RL

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7. Multi-Modal AI

Used for: Models combining text, image, audio, video

- **CLIP (OpenAI)** – Image-text similarity
- **OpenCLIP** – Open-source CLIP variant
- **FLAVA (Facebook)** – Multi-modal understanding
- **ALIGN (Google)** – Multi-modal image-text learning
- **MMF (Facebook)** – Multi-modal frameworks

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8. Specialized AI

Used for: Domain-specific AI

- **AlphaFold (DeepMind)** – Protein structure prediction
- **DeepMind Lab** – 3D RL and AI experiments
- **FastAI** (Python) – Simplified DL for vision/NLP/tabular data
- **OpenPose** (Python, C++) – Human pose estimation
- **SpeechBrain** (Python) – Speech recognition
- **DeepSpeech** (Mozilla, Python/C++) – Speech-to-text
- **Kaldi** (C++) – Speech recognition
- **PyCaret** (Python) – Low-code ML/AI

✅ **Tips:**

- Most Python libraries (TensorFlow, PyTorch, HuggingFace) cover multiple AI domains.
- Java-focused AI uses **DL4J, Weka, OpenNLP, and CoreNLP**, but Python dominates in AI.

If you want, I can make a **single massive table of 100+ AI libraries**, with **AI type, language, use-case, and link**, so you can see everything at a glance.

Do you want me to do that?

AI Libraries

```
├── 1. Machine Learning (ML)
│   ├── Scikit-learn
│   ├── XGBoost
│   ├── LightGBM
│   ├── CatBoost
│   ├── H2O.ai
│   ├── MLlib
│   ├── Weka
│   ├── Orange3
│   └── PyCaret
│
├── 2. Deep Learning (DL)
│   ├── TensorFlow
│   ├── Keras
│   ├── PyTorch
│   ├── MXNet
│   ├── Chainer
│   ├── Theano
│   ├── DL4J
│   ├── PyTorch Lightning
│   ├── FastAI
│   └── JAX + Haiku + Optax
│
├── 3. Natural Language Processing (NLP)
│   ├── NLTK
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│   ├── spaCy
│   ├── Gensim
│   ├── Transformers (HuggingFace)
│   ├── AllenNLP
│   ├── Stanford CoreNLP
│   ├── OpenNLP
│   ├── TextBlob
│   ├── Flair
│   └── Megatron-LM
│
├── 4. Computer Vision (CV)
│   ├── OpenCV
│   ├── Pillow/PIL
│   ├── scikit-image
│   ├── Detectron2
│   ├── YOLO (Ultralytics)
│   ├── MediaPipe
│   ├── OpenPose
│   ├── SimpleCV
│   ├── Tesseract OCR
│   ├── Open3D
│   ├── PyTorch3D
│   ├── Detecto
│   ├── ImageAI
│   ├── Albumentations
│   ├── OpenMMLab / MMDetection / MMAction2
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│   └── Kornia
│
├── 5. Generative AI
│   ├── GANs (TensorFlow / PyTorch)
│   ├── StyleGAN / StyleGAN2
│   ├── DALL·E / OpenAI API
│   ├── Stable Diffusion
│   ├── VQ-VAE / VQGAN
│   ├── MusicVAE
│   ├── DeepDream
│   ├── Diffusers
│   └── OpenAI Codex
│
├── 6. Reinforcement Learning (RL)
│   ├── OpenAI Gym
│   ├── Gym Retro
│   ├── Stable Baselines3
│   ├── RLlib
│   ├── TensorForce
│   ├── Dopamine
│   ├── Keras-RL
│   ├── Unity ML-Agents
│   ├── PettingZoo
│   ├── RLCard
│   ├── Coach
│   └── Acme
```

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│
├── 7. Multi-Modal AI
│   ├── CLIP
│   ├── OpenCLIP
│   ├── FLAVA
│   ├── ALIGN
│   └── MMF
│
├── 8. Specialized AI
│   ├── AlphaFold
│   ├── DeepMind Lab
│   ├── DeepChem
│   ├── RDKit
│   ├── SpeechBrain
│   ├── DeepSpeech
│   ├── Kaldi
│   ├── OpenAI Whisper
│   └── PyTorch BigGraph
│
└── 9. AutoML / Low-Code AI
    ├── Ludwig
    ├── AutoGluon
    ├── PyCaret
    └── FastAI
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